FLEET TEAM DIGEST

737MAX-FTD-32-19002

Issue Title : NLG Shock Strut Fluid Leakage

Airplane Model	ATA	Minor Model(s)
737MAX	3221-11	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
02/11/2020	04/27/2020	04/20/2020

Estimated Completion Date	Next Update Date

Revision Description

Milestones Section revised to reflect CMM release date.

FTD now considered closed.

Applicability

All 737MAX Airplanes.

Description

Several 737MAX operators have reported finding indications of Nose Landing Gear (NLG) shock strut fluid leakage while the airplane is parked. Boeing opened an In-Service Investigation (ISI) to review the design and investigate for a potentional quality escapement.

Background

As part of the investigation, Boeing also found several 737MAX airplanes located at Boeing's delivery center exhibiting NLG shock strut fluid leakage. Same as the other reports, the fluid was observed leaking from the weep hole located at the bottom of the inner cylinder.

It was determined the leakage was originating between the inner cylinder assembly and the metering pin. A detail examination on the inner cylinder assembly weep hole via use of a borescope found that small drops were originating from the seal installed on the metering pin (AIPC 32-21-00-01, item 350). Photographs are provided with this FTD article for reference. Boeing issued a quality investigation to determine the cause of the leakage across the metering pin seal.

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Status

The Boeing investigation found the chrome plating surface finish was not tightly controlled in the seal groove area of the inner cylinder during manufacturing. This may allow fluid to slowly by-pass the seal, thus resulting in fluid indications to appear at the bottom of the inner cylinder weep hole. The leak is very slow and would not fail the Acceptance Testing Procedure performed on every strut, nor a high pressure test that was done on one of the leaking struts after discovery of the issue.

Boeing also reviewed the seal design, and determined that improving the seal design can also reduce the likelihood of leakage.

Boeing has revised the engineering drawing requirements to control the inner cylinder chrome plating surface finish, and use an improved seal, P/N 723K2MTE-161-P8. These improvements are introduced in NLG Component Installation, P/N 162A1100-304 (for the 737MAX-7/-8/-8200/-9) and P/N 162A1100-305 (for the 737MAX-10).

Interim Action

If operators are experiencing 737MAX NLG fluid leakage through the metering pin seal, Boeing recommends performing Aircraft Maintenance Manual (AMM) task 32-21-11-200-802, 'Nose Landing Gear Shock Strut Metering Pin Seal Leakage Check'. This check will ensure the NLG fluid leakage is within an acceptable leak rate.

If excess fluid leakage is suspected, Boeing recommends performing a two point pressure check per AMM task 12-15-41-610-801, 'Nose Landing Gear Shock Strut Fluid Check', to ensure that the correct fluid volume is present within the NLG. Check the pressure and X-dimension on each strut at two different aircraft weights (no passengers and low fuel, then with some fuel and/or cargo/passengers). The pressure and X-dimension must fall on the servicing chart at both aircraft weights. If the NLG X-dimension cannot land within the servicing chart, the NLG will required to be serviced (shock strut fluid and nitrogen gas) per AMM task 12-15-41-610-802, 'Nose Landing Gear Shock Strut Servicing, Airplane on the Ground'.

Final Action

Boeing recommends operators to service the NLG every 3000 flight cycles per 737MAX Maintenance Planning Data (MPD) item number 32-070-00. Operators are encouraged to use this NLG servicing frequency or adjusted at/below the recommended value based on operator experience.

Boeing incorporated the following design changes to mitigate these events from re-occurring in the 737MAX fleet. NLG shock struts on Line Numbers 7765 - 7886 have inner cylinders with the improved seal groove chrome plating surface finish only. NLG shock struts on Line numbers 7887 – 7934 (NLG Component Installation P/N 162A1100-303) were retroactively revised with the improved chrome plate finish and new seal installation. A final design change controls the chrome plating finish on the inner cylinder and installs the new seal starting in line number 7935 and on (NLG Component Installation P/N 162A1100-304 / -305).

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Boeing will revise Component Maintenance Manual (CMM) 32-21-12 to reflect the new metering pin seal configuration, as well as the inner cylinder seal groove finish requirements. The CMM will also list the improved seal part numbers as the preferred optional seal for the previous NLG shock strut configurations, thus allowing retrofit of the improved seals at landing gear overhaul.

Boeing will revise the Airplane Illustrated Parts Catalog (AIPC) 32-21-00-XX to reflect the new metering pin seal configuration part numbers. (Note: The part number of the inner cylinder does not change.)

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Production Incorporation: LN 7935 (see Final Action for further details on phased production incorporation)

CMM Incorporation: CMM 32-21-12 Rev 58 dated 01-MAR-2020.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Component Maintenance Manual (CMM)	32-21-12		
Illustrated Parts Catalog (IPC)	32-21-00-01 item 350		
Aircraft Maintenance Manual (AMM)	32-21-11-200-802		
Production Revision Record (PRR)	95000-2945		08/06/2019

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Inner Cylinder Assy	162A1120-301		
Metering Pin Subtitution Kit Assy	162A1100-350		
NLG Component Installation	162A1100-301		
NLG Component Installation	162A1100-302		
NLG Component Installation	162A1100-303		
Metering Pin Seal Assy	723K2MTE-161-4551	Green Tweed & Co.	5F573
Metering Pin Seal Assy	7232MT-160-P8	Green Tweed & Co.	5F573

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Metering Pin Seal Assy	S34702-232BAK29	Trelleborg Sealing Solutions	09257

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Manufacturing Quality Issues : Issues related to airplane or component manufacturing quality

Attachments:

a) 737MAX NLG Weep Hole Leakage Findings

b) NLG Inner Cylinder Weep Hole Location

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